Analyzing PC-ACE Access Data in the NLP Suite

QNA (Quantitative Narrative Analysis)

In a book chapter published in 2012 (Franzosi, 2012a: 83-84), Franzosi succinctly summarizes the steps involved in carrying out Quantitative Narrative Analysis (QNA).

> In the early 1980s, in search of the actor in the study of social protest and violence, I started working on a new methodological approach focused on social actors rather than variables. Key to this approach was the idea of taking stories of conflict, coming mostly from newspaper articles, and turning their words into numbers. Indeed, for a time, several of my publications bore the title "From Words to Numbers" for the technique (Franzosi, 1989, 1994, 2004). Then, I settled on Quantitative Narrative Analysis for its easy acronym, QNA (Franzosi, 2010). QNA involves four steps:
>
> 1. Identify and assemble a set of narrative documents for analysis (e.g., newspaper articles of conflict and violence; but could also be police reports, personal narratives as told in in-depth interviews, web narratives, or fictional narratives).
> 2. Design a "story grammar" (i.e., a basic linguistic structure that would capture in a set of categories the information contained in narrative documents namely, information on actors, their actions, and the attributes of actors and actions).

3. Design a computer program to store both the grammar and the coded information so as to make possible large-scale socio-historical projects (PC-ACE, Program for Computer-Assisted Coding of Events);
4. Analyze the data with statistical tools most appropriate for the narrative nature of the data (i.e., actors acting in time and space, rather than variables; for example, network models or GIS [Geographic Information System] tools, instead of regression models).

Franzosi's book *Quantitative Narrative Analysis* (2010: 3, 4) expands on the linguistics properties of the new technique.

> The tradition this book draws on is very different from the quantitative approach to texts that has been dominant in the social sciences: content analysis (Franzosi, 2004a, 2008; Krippendorf, 1980; Weber, 1990). The fundamental difference between quantitative narrative analysis (QNA) and content analysis, in its multiple variants, is that the categories of QNA are based on linguistic properties of texts—narrative texts in particular—as drawn from a literary and linguistic tradition that has worked long and hard at teasing out those properties. Content analysis, on the other hand, bases its categories on the investigator's theoretical and/or substantive interests. In content analysis, categories vary from investigator to investigator, from subfield to subfield, although some sedimentation of categories is inevitable *within* subfields as new investigators address previous scholarly work. … What content analysis and quantitative narrative analysis have in common, however, is the drive to quantify, to use words for the purpose of extracting numbers out of them, something foreign to the linguistic tradition of narrative analysis. The invariant linguistic properties of narrative on which QNA is based, however, help deliver more hypothesis-free data than content analysis, with its coding schemes based on investigators' theoretical interests.

PC-ACE (Program for Computer-Assisted Coding of Events)

Franzosi continues (2010: 59, 61, 76, 102):

[W]ithout software, quantitative narrative analysis is not feasible, at least not on a large scale (consider the 18,000 documents of my project on Italian fascism analyzed in this book). The sheer complexity of the grammar in terms of the number of categories and their relational properties would confine the use of QNA to trivial examples. If "no software, no QNA," what are the options? … PC-ACE probably offers the most comprehensive approach to quantitative narrative analysis, but CAQDAS packages as well do allow a minimal implementation of QNA. … PC-ACE (Program for Computer-Assisted Coding of Events, available for free download at http://www.pc-ace.com) is a software program that I designed to carry out quantitative narrative analysis for large-scale sociohistorical research (for a more in-depth, technical treatment, see Franzosi & Cunial, 2009). It is a Microsoft ACCESS application designed to deal with complex data structures in very flexible ways. … PC-ACE (Program for Computer-Assisted Coding of Events, available for free download at http://www.pc-ace.com) is a relational database management system (RDBMS) application of Microsoft Access that I designed to carry out quantitative narrative analysis. PC-ACE has several features designed for optimal quantitative narrative analysis (e.g., setup of a specific story grammar, maximization of data-entry speed and data reliability, data query and data update).[1]

Figure 1 – The opening window of PC-ACE (Program for Computer-Assisted Coding of Events)

The combination of QNA and PC-ACE allowed users to collect far richer, more reliable, and nearly hypothesis-free event data, where someone does something in time and space pro/against someone else. PC-ACE displayed data in narrative form (see below, where the setup categories of the data collection grammar are in black and the data taken from source documents (typically, newspapers) are in red.

Semantic triplet: [Subject: [Actor... (Name of individual actor: negro)]] [Action: (Verb phrase: attacks from behind) [Time... (Definite date: October 14, 1896)] [Space..City: (City name: near Griffin) (Locality: woods)]]] [Object: [Actor... (Name of individual actor: woman)] [Name of individual: Blanche Gray] [Marital status: single]]

Semantic triplet: [Subject: [Actor... (Name of individual actor: negro)]] [Action: (Verb phrase: chokes) [Time... (Definite date: October 14, 1896)] [Space..City: (City name: near Griffin) (Locality: woods)]]] [Object: [Actor... (Name of individual actor: woman)] [Name of individual: Blanche Gray] [Marital status: single]]

Semantic triplet: [Subject: [Actor... (Name of individual actor: negro)]] [Action: (Verb phrase: rapes) [Time... (Definite date: October 14, 1896)] [Space..City: (City name: near Griffin) (Locality: woods)]]] [Object: [Actor... (Name of individual actor: woman)] [Name of individual: Blanche Gray] [Marital status: single]]

Franzosi used PC-ACE on two different large-scale socio-historical projects: the rise of Italian fascism (1919-1922) (5 newspapers and over 50,000 semantic triplets) and lynchings in Georgia (1875-1930) (over 5,0000 newspapers and 7,000 semantic triplets). In this chapter, as a way of illustration, we focus on the Georgia lynchings project.

Unfortunately, PC-ACE is no longer supported and the domain name (www.pc-ace.com) has been lost. In recent years, Franzosi has shifted his focus on developing automatic, rather than computer-assisted, approaches to textual analysis based on Natural Language Processing (NLP) tools: the NLP Suite, an open-source, freeware Python package publicly available on GitHub (https://github.com/NLP-Suite/NLP-Suite/wiki) (see Fig. 2).

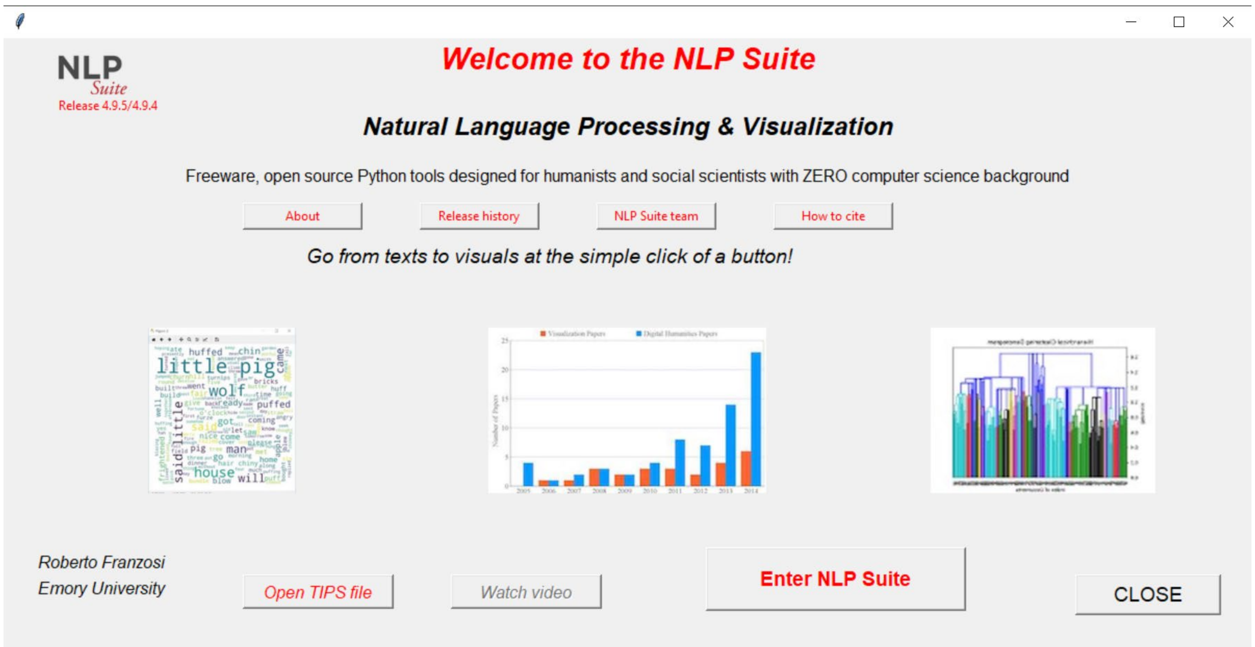

Figure 2 – The opening GUI (Graphical User Interface) of the NLP Suite

The NLP Suite contains not just NLP tools (annotators, parsers, coreference resolution, sentiment analysis in BERT, spaCy, Stanford CoreNLP, Stanza), but a variety of data handling and data visualization tools (e.g., standard charts such as bar, line, pie, charts, but also wordclouds, sunburst, treemap, colormap, and Sankey charts, network graphs, geographic maps in Google Earth Pro and Google Map and Python folium maps).

Among the Data & Files Handling Tools available in the NLP Suite main menu GUI (Graphical User Interface) (Fig. 3), there is a set of functions to analyze and visualize the database data collected with PC-ACE (via Pandas).

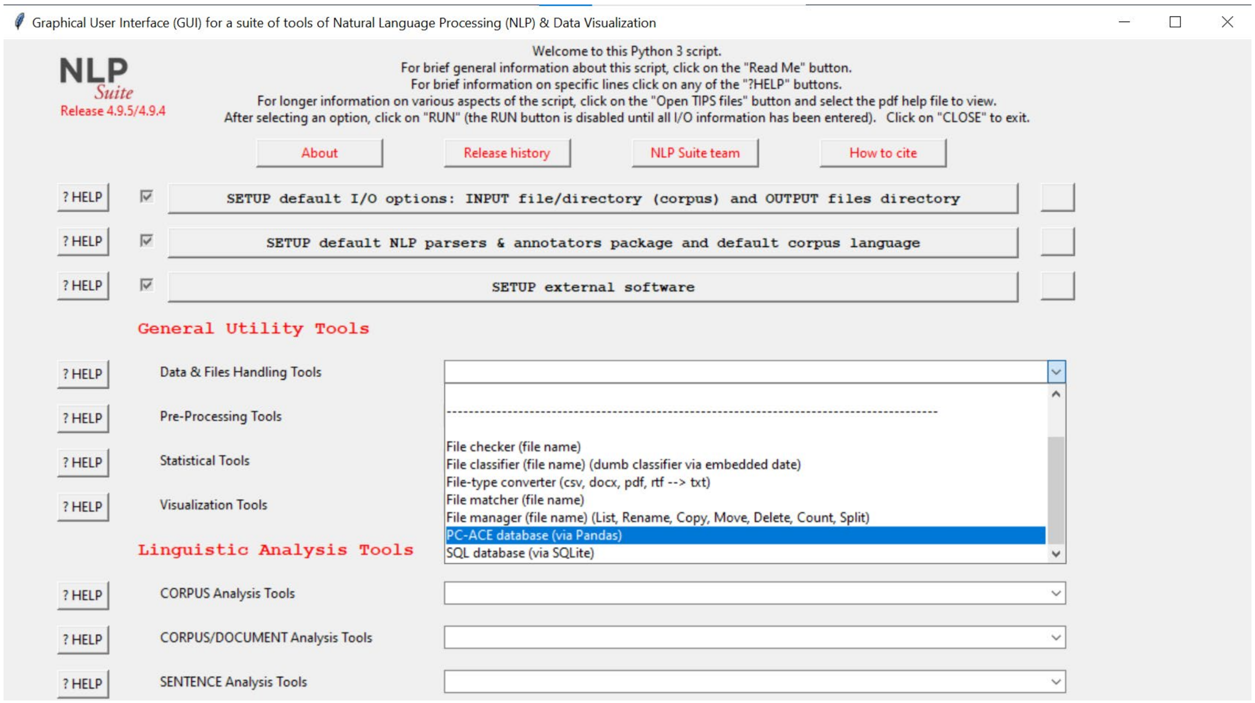

Figure 3 – The main menu GUI (Graphical User Interface) of the NLP Suite

The DB_PCACE_data_analyzer GUI

Being built on a Microsoft ACCESS relational database, “you can use PC-ACE for a bibliography, a telephone directory, a company-s parts or customer inventory,” but, of course, the best application is narrative analysis (Franzosi, 2010: 76). The DB_PCACE_data_analyzer GUI was designed to allow NLP Suite users to analyze narrative data previously collected using PC-ACE (Program for Computer-Assisted Coding of Events). And since QNA is fundamentally about social actors doing social actions in time and space, the 5 Ws of journalism, Who, What, When, Where, and Why, the NLP Suite PC-ACE GUI should allow users to analyze any of these items and, especially, the relationship between the Who and the What.

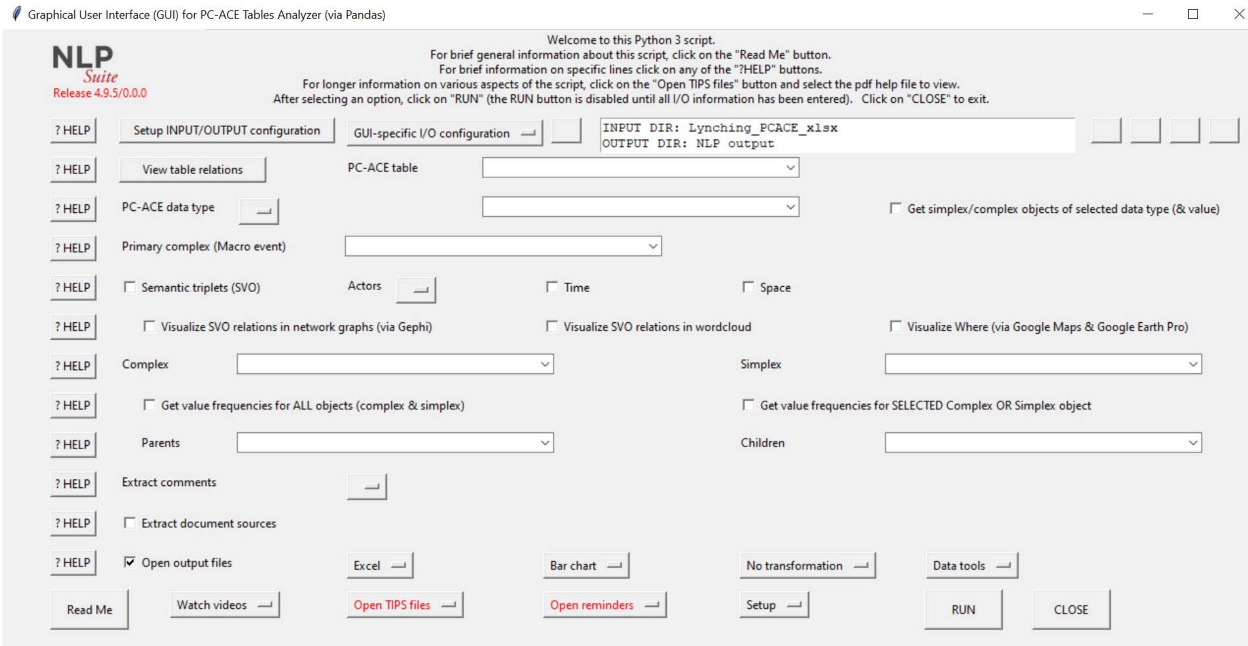

Figure 4 – The PC-ACE Table Analyzer GUI (Graphical User Interface) of the NLP Suite

The widgets in the GUI allow the user to explore different aspects of the data collected. Users can explore the setup structure of the grammar, finding parents and children of any object. Users can obtain frequency distributions and bar charts of any complex and simplex object. Social actors involved in the events (namely, individuals, collective actors, or organizations), time and space. Fundamentally, the GUI allows users to export the semantic triplets (or Subject-Verb-Object) coded in the PC-ACE database and visualize the results in network graphs.

Actors

Fig. 5 and Fig. 6 provide the frequency distributions of the most frequent social actors (individual and collective) found in the Georgia lynching PC-ACE database (1875-1935). Among the individuals, negros, in the newspaper language of the times, are the most frequent actors at 3 times the frequency of white men. White women, women, and girts also appear as the most frequent individual actors. Some 50% of all lynchings occur over violence of gender norms where even looking at a white woman may be a violation of such norms. And mobs and posse are among the most frequent collective actors (Fig. 6) along with men (white men) and crowds.

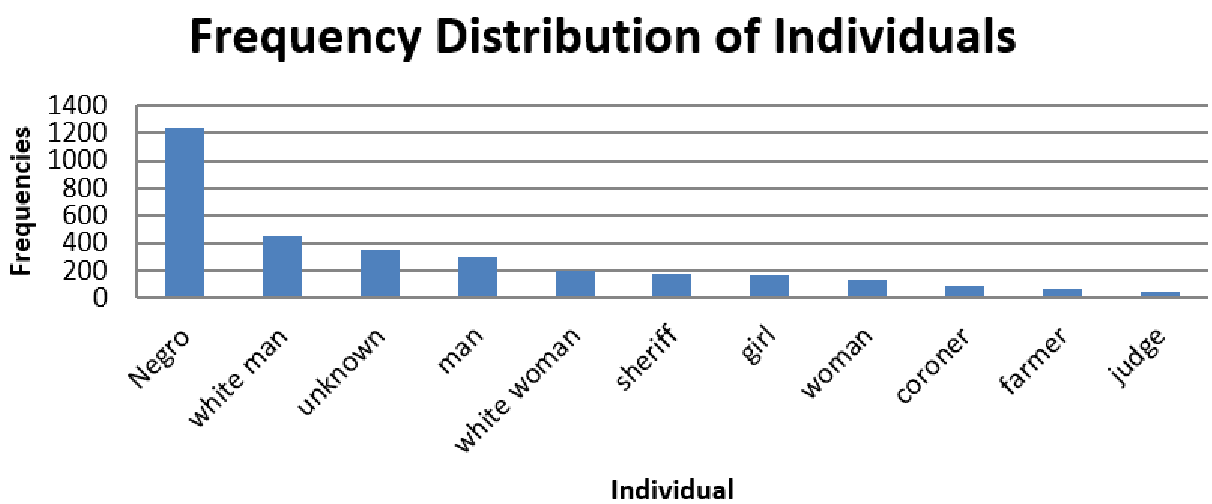

Figure 5 – Bar chart of the most frequent individuals mentioned in the Georgia lynching (1875-1935) PC-ACE database

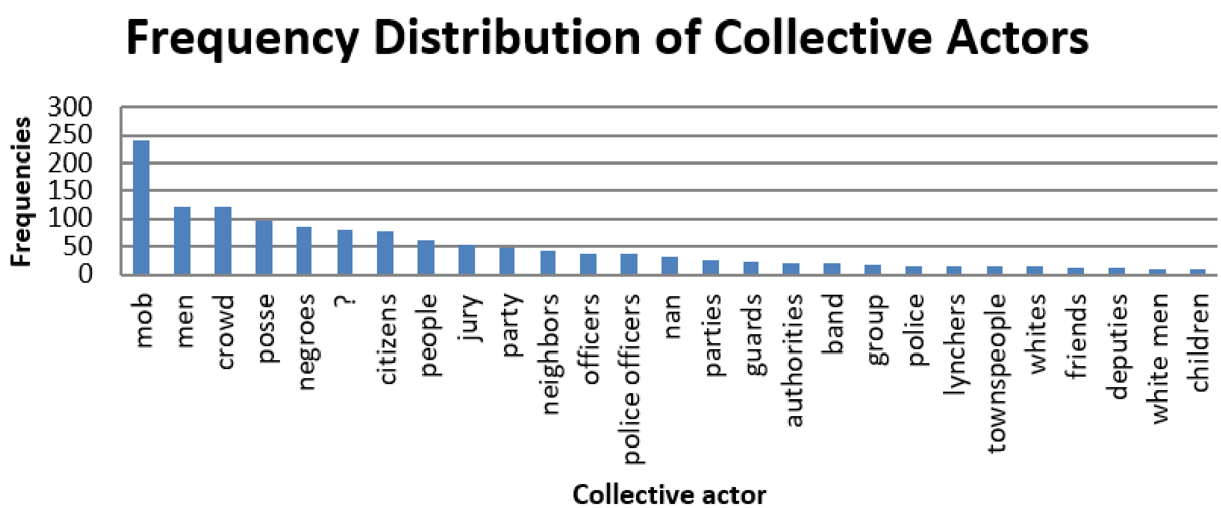

Figure 6 – Bar chart of the most frequent collective actors mentioned in the Georgia lynching (1875-1935) PC-ACE database

Time of day

Figure 7 displays a bar chart of the most frequent times of day (with more than 100 mentions) in the Georgia lynching (1875-1935) PC-ACE database. Most actions occur at night. Certainly, lynchings as the SVO results of Who does What would confirm.

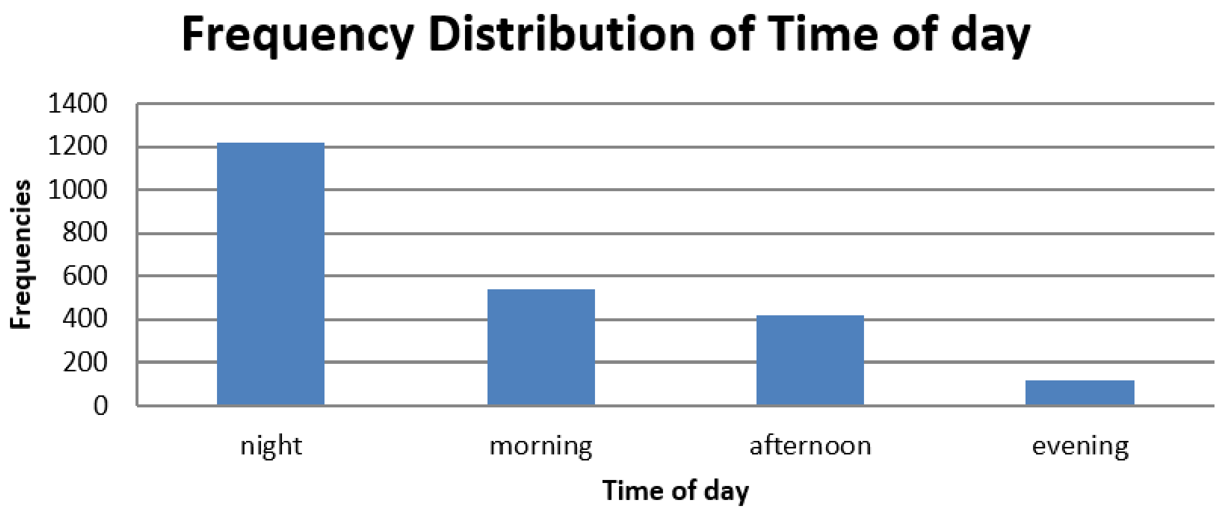

Figure 7 – Bar chart of the most frequent times of day mentioned in the Georgia lynching (1875-1935) PC-ACE database

Space

Thea Space widget in the PC-ACE tables analyzer GUI extracts and visualizes all locations mentioned in a PC-ACE database. In the Georgia lynchings PC-ACE database, there are over 300 locations. Most locations have very low frequencies (in fact, a frequency of 1 for some 50 locations), but some, like Macon, Athens, East Point have very high frequencies (see Fig. 8).

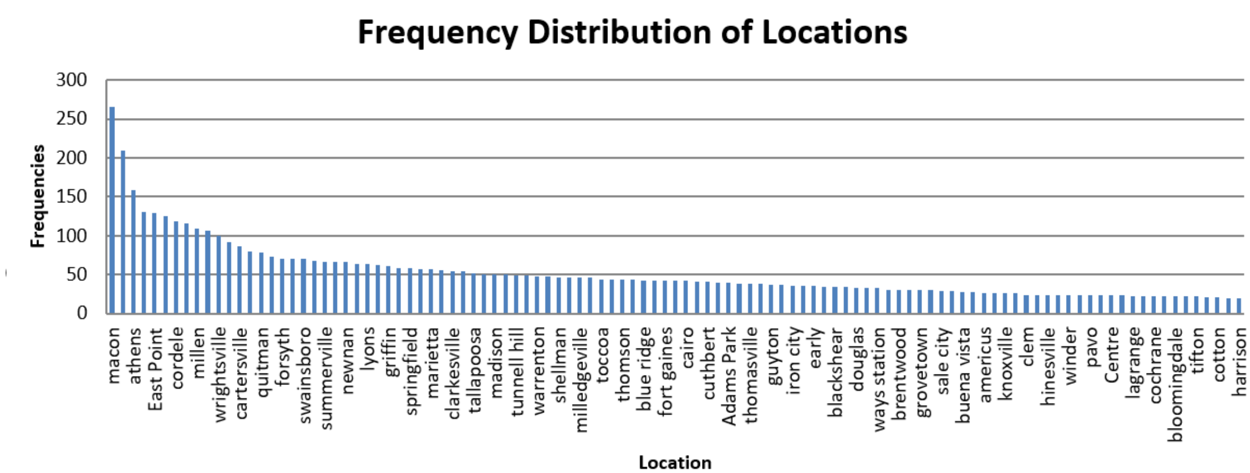

Figure 8 – Bar chart of locations mentioned in the Georgia lynching (1875-1935) PC-ACE database (with frequencies greater than 20)

There are also several mentions of countries in the database (see, Fig. 9) with 6,000 references to the United States and Canada trailing at a distant 176 frequency.

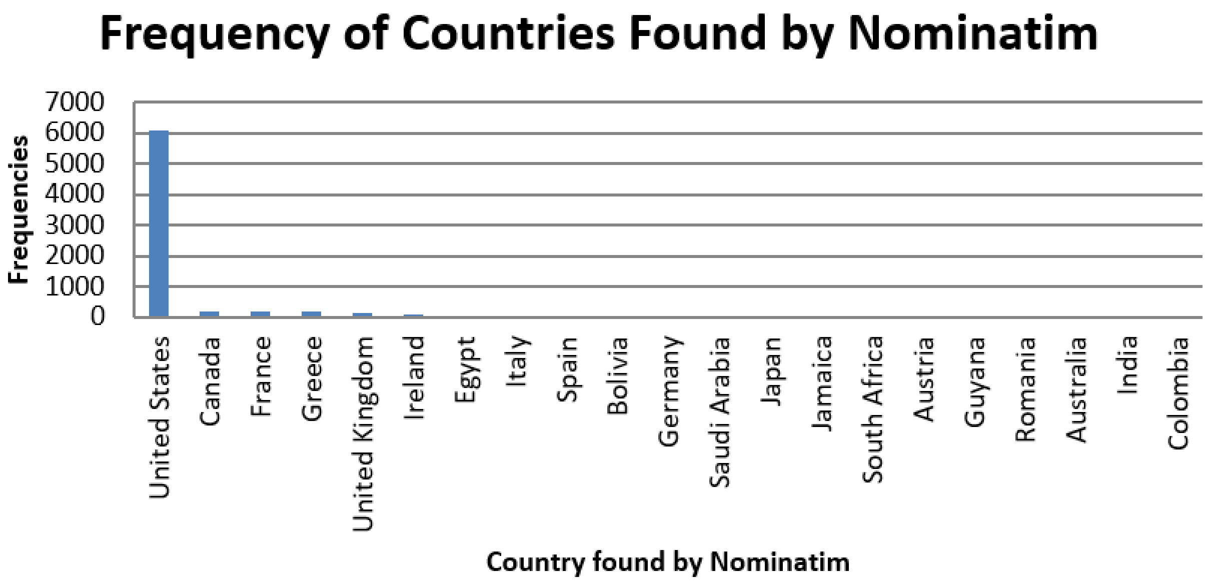

Figure 9 – Most frequent countries mentioned in the Georgia lynching (1875-1935) PC-ACE database (with frequencies greater than 5)

The distinct advantage of working with the NLP Suite is that the Space algorithms can take advantage of several functions available in the Suite: geocoding of location names via Nominatim (i.e., getting the latitude and longitude waypoints indispensable for mapping) and mapping in Google Earth Pro (pin map, Fig. 10) and Google Maps (heat map, Fig. 11).

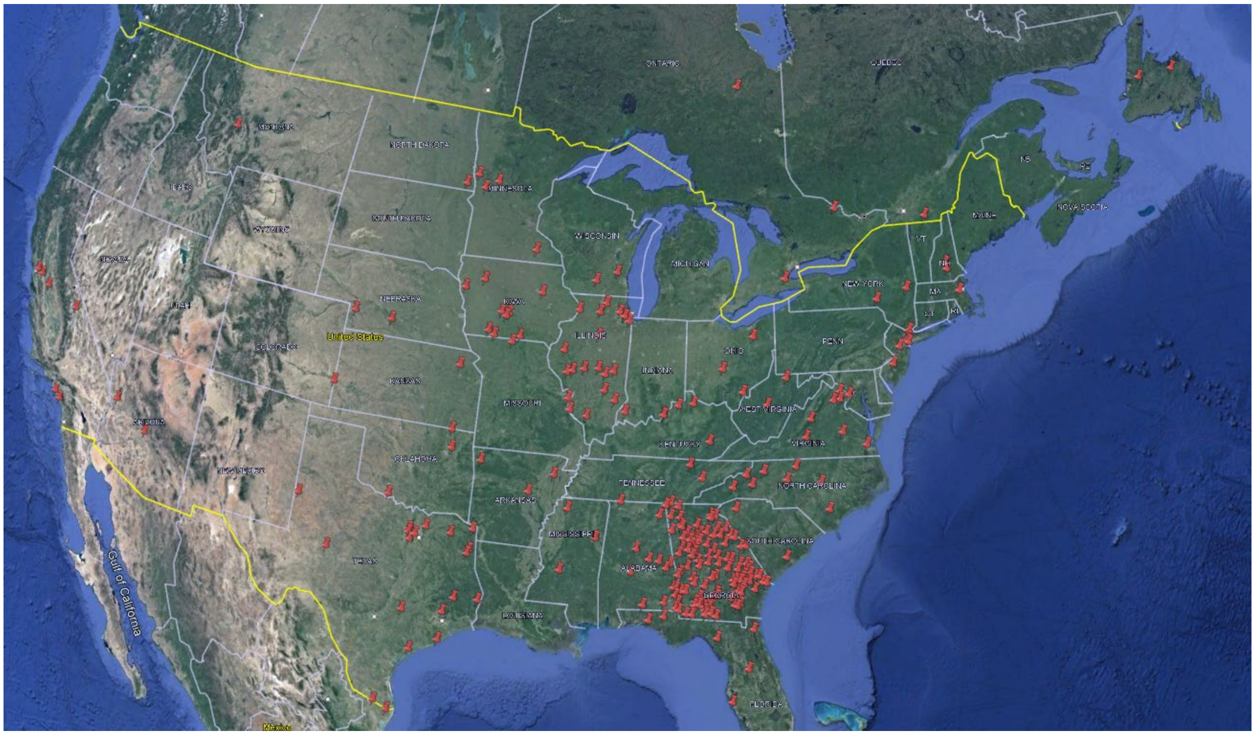

Figure 10 – Google Earth Pro pin map of locations mentioned in the Georgia lynching (1875-1935) PC-ACE database

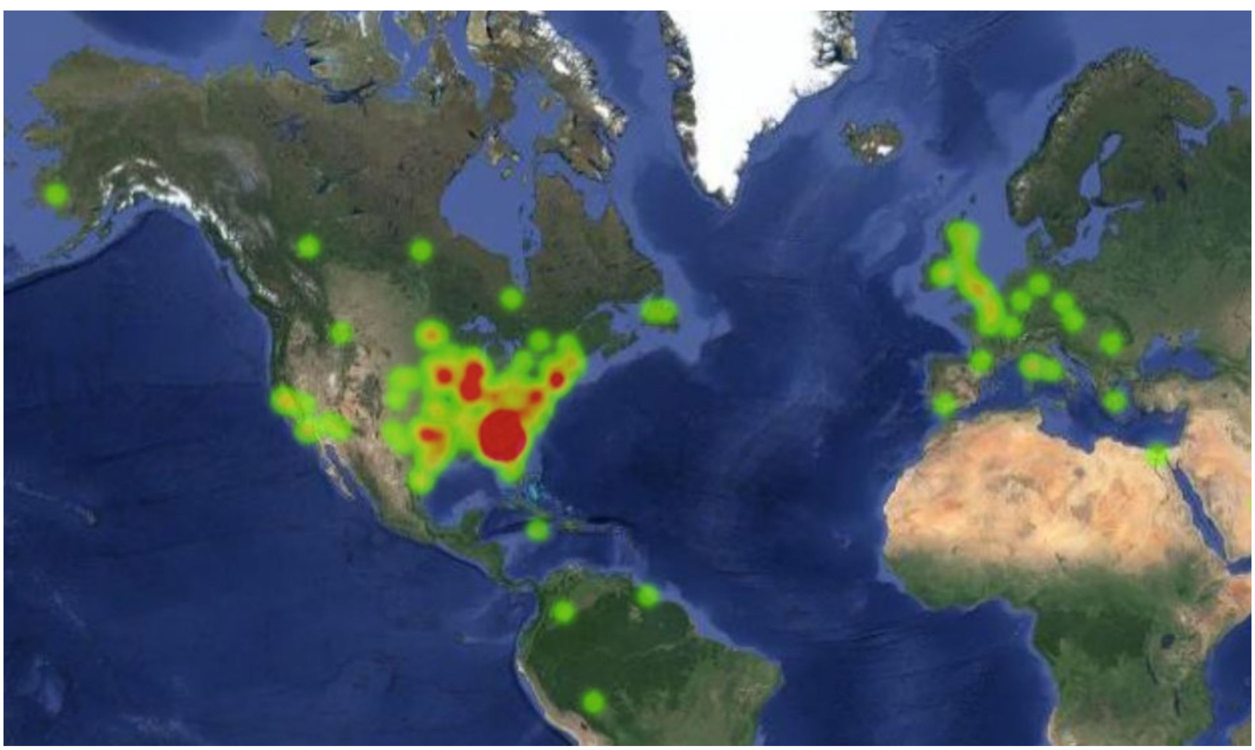

Figure 11 – Google Maps heat map of locations mentioned in the Georgia lynching (1875-1935) PC-ACE database

Both pin and heat maps show how far-reaching where the news of lynching events.

Semantic Triplet: SVO

The semantic triplet (or SVO, Subject, Verb, Object) is at the heart of QNA (see Franzosi, 2010: 25-27). The Semantic triple (SVO) widget in the GUI not only allows to extract all the triplets available in a PC-ACE database but also to visualize the results in Sankey flowcharts, Gephi network graphs, and wordclouds. Sankey and Gephi charts are interactive: by hovering over any actor, the chart will highlight the action (Verb) and Object of the action for the selected actor. Unfortunately, in print, these interactive effects are lost. The Sankey flowchart of Fig. 12 visualizes the top 5 most frequent Subjects (sheriff, crowd, mob, posse, negro, in the newspaper language of the time). The Gephi network graphs of Fig. 13 and Fig. 14 (with the actor mob highlighted) display the wide set of relations coded as semantic triplets.

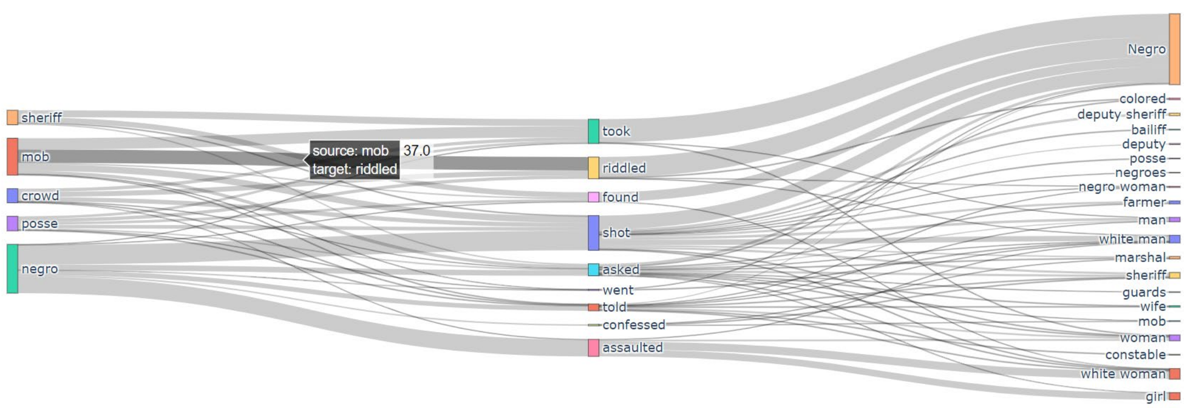

Figure 12 – Sankey flowchart of semantic triplets (SVOs – Subject-Verb-Object) relationships mentioned in the Georgia lynching (1875-1935) PC-ACE database

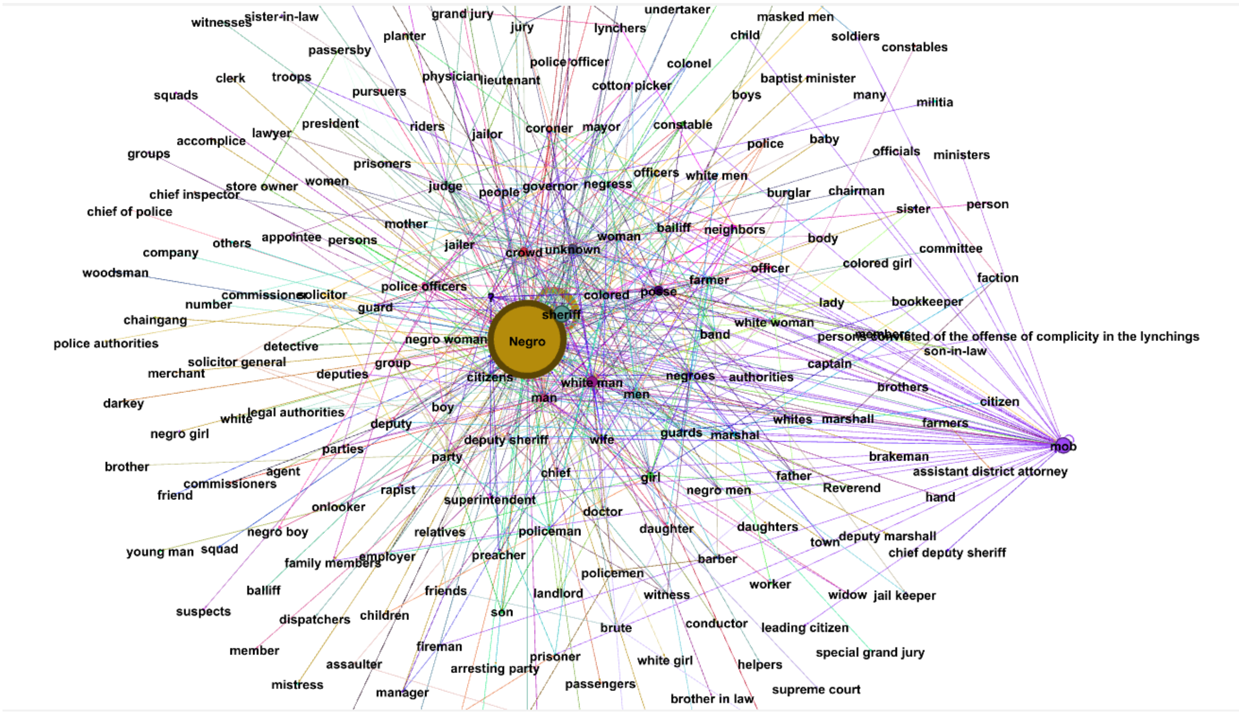

Figure 13 – Gephi network graph of semantic triplets (SVOs – Subject-Verb-Object) relationships mentioned in the Georgia lynching (1875-1935) PC-ACE database

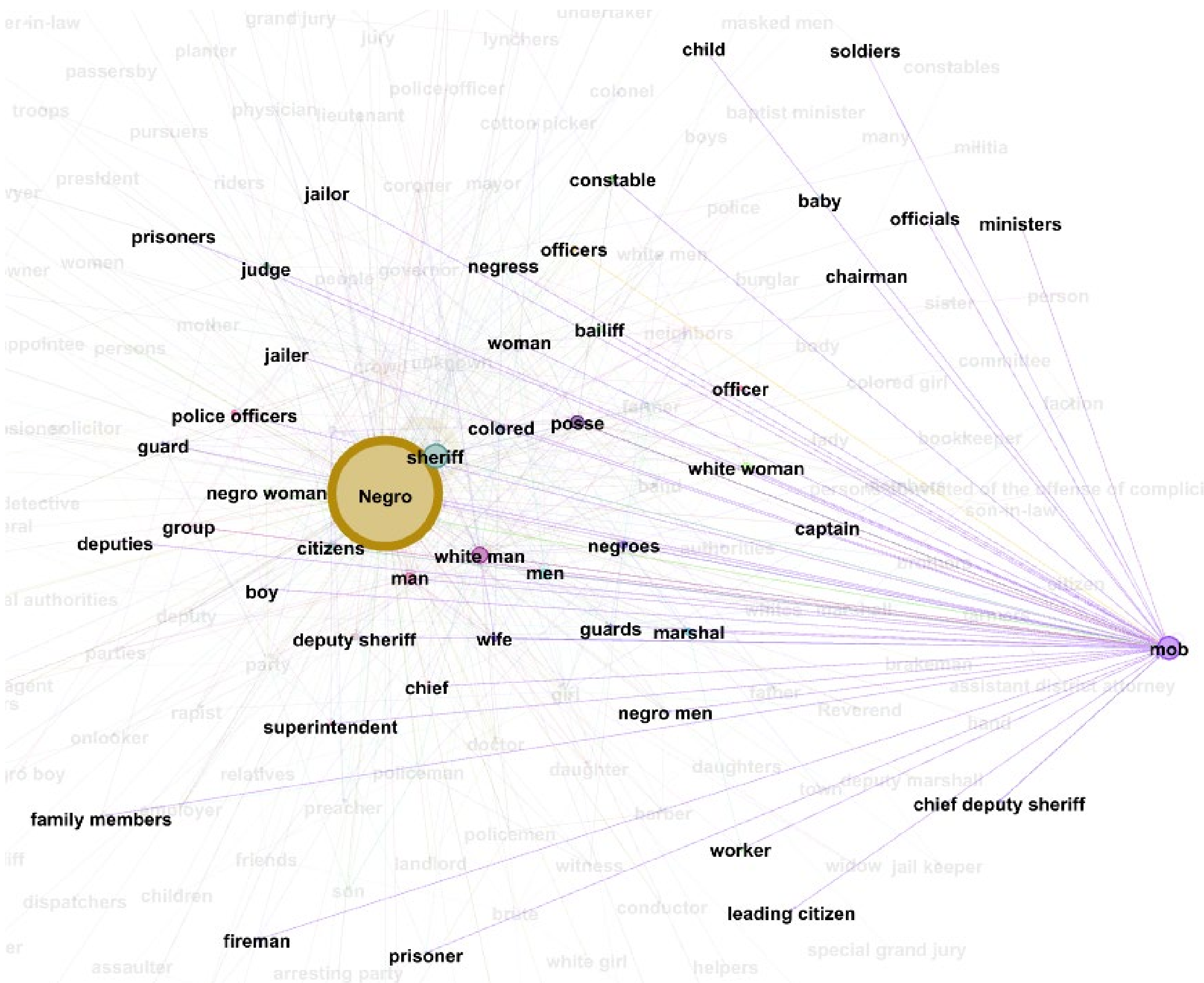

Figure 14 – Gephi network graph of semantic triplets (SVOs – Subject-Verb-Object) relationships mentioned in the Georgia lynching (1875-1935) PC-ACE database (hovering over mob)

The wordcloud of Fig. 15 provides a succinct summary of all semantic triplets, with Subjects in red, Verbs in blue, and Objects in green.

Figure 15 – Wordcloud of semantic triplets (SVOs – Subject-Verb-Object) relationships mentioned in the Georgia lynching (1875-1935) PC-ACE database: Subject (red), Verb (blue), Object (green)

Conclusions

In this chapter, we have shown how to convert the tables of a PC-ACE Microsoft ACCESS database into Python Pandas files, extract the information, and analyze it with a variety of statistical and visualizations tools available in the Python package NLP Suite, publicly available on GitHub (https://github.com/NLP-Suite/NLP-Suite/wiki). The question is: Why bother? There are two good reasons.

First, PC-ACE is no longer available and supported. But you may have data stored in a PC-ACE database that you need to analyze. PC-ACE, unfortunately, has very limited data analysis and visualization options. To carry out more sophisticated analyses beyond frequency distributions, you need to be able to export your database tables so that you can analyze the data in a different environment.

Second, the NLP Suite provides such analytical environment for PC-ACE data.[2] The NLP Suite offers a wide range of data analysis options well beyond the ones shown here. You can analyze and visualize data with Sunburst, treemap, colormap, timemapper charts, boxplots. Furthermore, if the source documents available in PC-ACE database are in text format, the package offers a wide range of automatic textual analysis options, from parsers to SVO extractor, automatic ways of going from texts to geographic maps via Stanford CoreNLP NER (Named Entity Recognition) to extract locations, geocode them via Nominatim, and map them via Google Earth Pro, Google Maps or Python Folium package, coreference resolution, sentiment analysis in BERT, spaCy, Stanford CoreNLP, and Stanza, style analysis with n-grams and co-occurrences, text readability and sentence complexity, knowledge graphs via WordNet, DBpedia, and YAGO.

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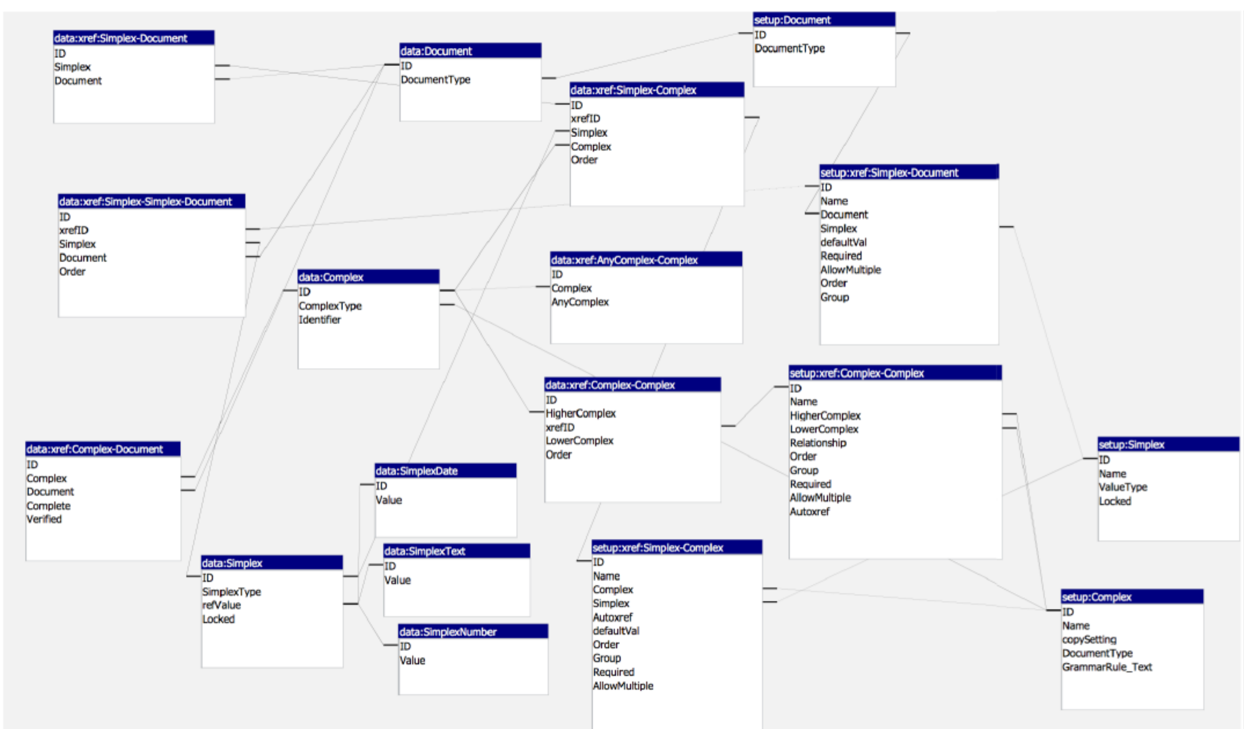
Appendix

PC-ACE ACCESS Table Relations

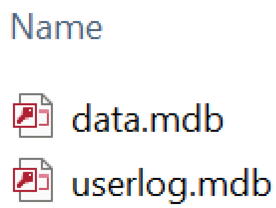
PC-ACE is based on a large set of relational tables. Table names and table field names are fixed in PC-ACE. Only the setup values of both complex and simplex objects change at the users' discretion, depending upon the complexity of the grammar they wish to setup.



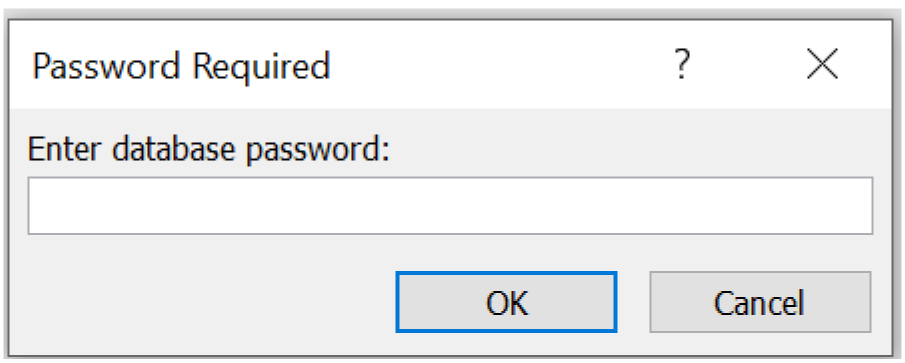
Export PC-ACE ACCESS Tables to Excel

All tables from both **data** and **userlog** backend databases will need to be exported to Excel.

Enter the database folder.



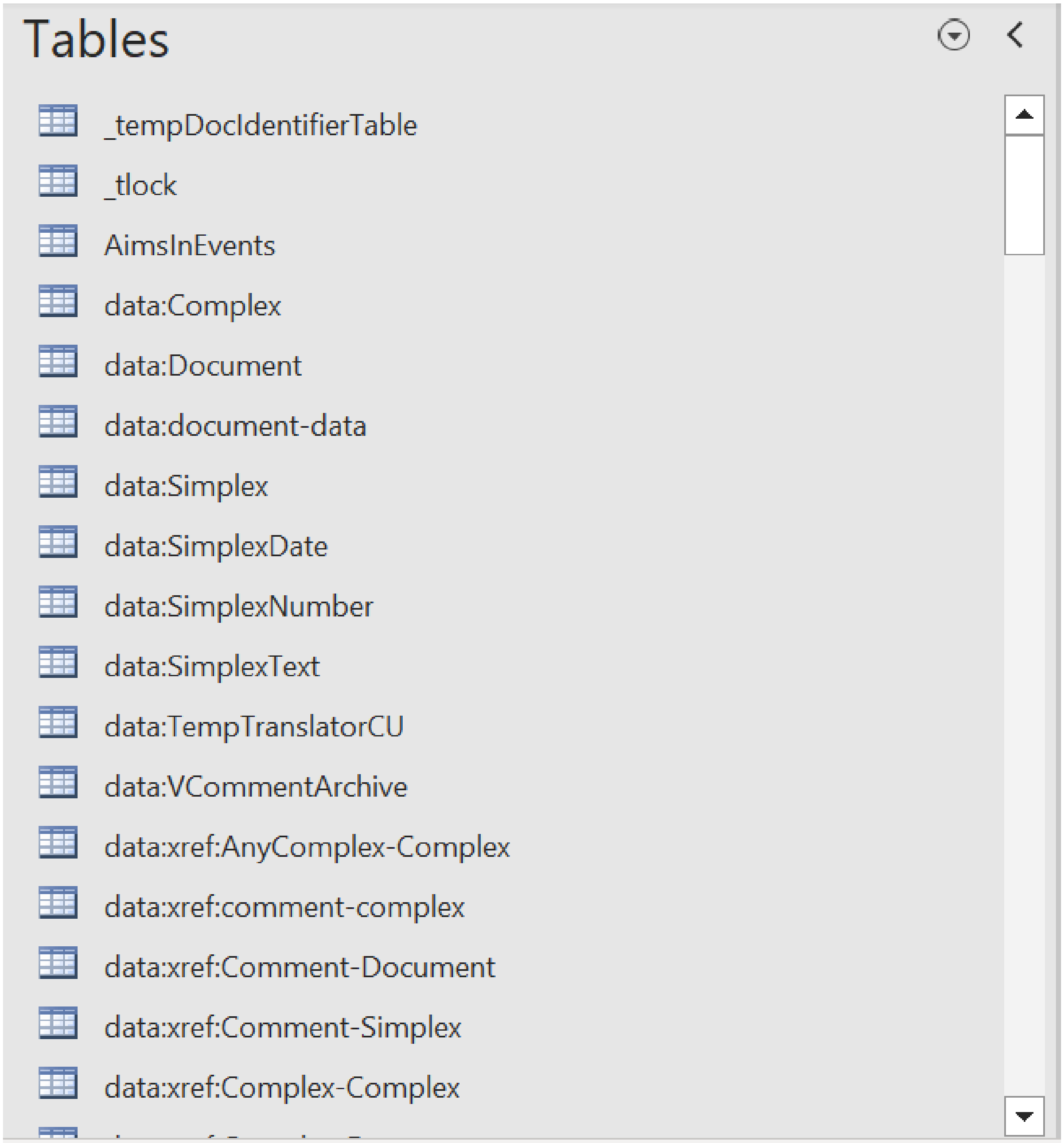
Double click on the data database. When prompted



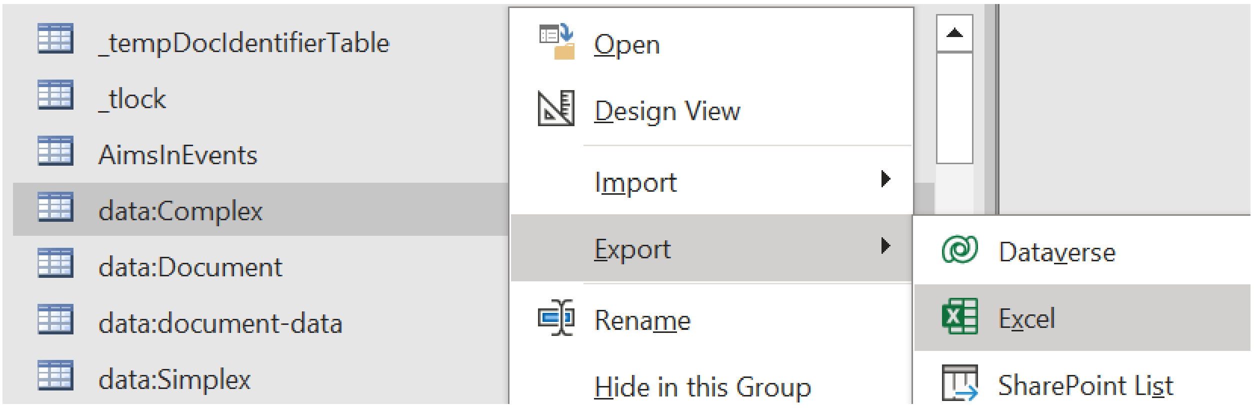
Enter the password *pcacenarrativedatabackend*

The password for the userlog database is *pcacenarrativedataba*

You will see a list of available tables.

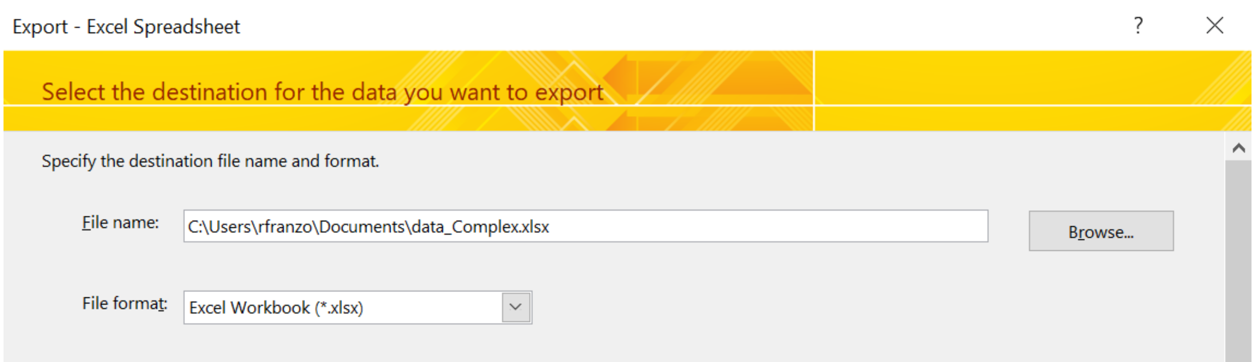


Double click on the first table to be exported: data:Complex.

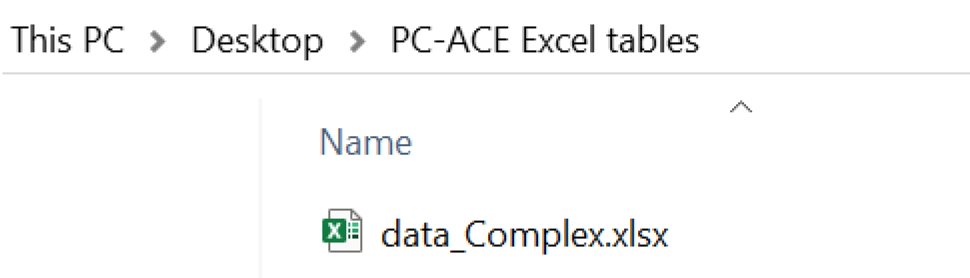


Select the Export option, and the Excel option.

Use the Browse button to select the folder where to export the Excel file then click OK.



The exported Excel file will now appear in the selected folder (e.g., PC-ACE Excel tables).



Do so for all the tables in both data and userlog databases.

Name

data_Complex.xlsx
data_Document.xlsx
data_document-data.xlsx
data_Simplex.xlsx
data_SimplexDate.xlsx
data_SimplexNumber.xlsx
data_SimplexText.xlsx
data_VCommentArchive.xlsx
data_xref_AnyComplex-Complex.xlsx
data_xref_comment-complex.xlsx
data_xref_Comment-Document.xlsx
data_xref_Comment-Simplex.xlsx
data_xref_Complex-Complex.xlsx
data_xref_Complex-Document.xlsx
data_xref_Simplex-Complex.xlsx
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data_xref_Simplex-Simplex-Document.xlsx
data_xref_VComment.xlsx
data_xref_VComment-Document.xlsx
setup_Complex.xlsx

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[1] On a comparison of QNA implementations in PC-ACE and various CAQDAS packages, see Franzosi et al. 2013.

[2] R, Stata, SPSS could also be suitable analytic options.